ABC Startup Solutions — Complete IT Infrastructure Plan

Planning baseline: 20 employees, one office floor, infrastructure built from zero.
Consistency decisions used throughout this plan:

- **14 desktop PCs + 6 laptops = 20 employee computers**
- **24 monitors** to support dual-monitor setups for IT and selected Finance/Sales workstations
- **1 Ubuntu Server**, **1 NAS**, and **2 external backup drives**
- **1 managed 48-port switch** with VLAN segmentation
- **2 wireless access points**
- **4 department VLANs:** IT, HR, Finance, Sales
- **1 infrastructure/server VLAN** for server, printer, and network management
- **Windows 11 Pro 25H2** for employee endpoints. Microsoft currently lists Windows 11 25H2 as a supported Pro release; 26H1 also exists, but 25H2 is selected here as the common deployment baseline for consistency and application compatibility.
- **Ubuntu Server 24.04.4 LTS** for the on-premises server. Ubuntu 24.04 LTS has long-term support, and the 24.04.4 point release was current as of February 2026.

Part 1 — Company Profile

1.1 Company Name

ABC Startup Solutions

1.2 Nature of Business

ABC Startup Solutions is a fictional small-to-medium software development company that designs, develops, tests, deploys, and maintains custom software applications for business clients.

The company provides:

- Web and application development
- Software testing and quality assurance
- Database and API development
- Technical consulting
- Software maintenance and support
- Cloud and systems integration services

Because software development is the core business, the IT infrastructure must support reliable development work, secure business information, remote access, collaboration, backups, and future growth.

1.3 Company Vision

To become a trusted technology partner that delivers innovative, secure, and dependable software solutions that help organizations operate and grow.

1.4 Company Mission

ABC Startup Solutions provides practical, high-quality software and technology services by combining skilled people, modern infrastructure, secure systems, and continuous improvement.

1.5 Fictional Office Location

ABC Startup Solutions
Unit 402, Innovation Tower
J.P. Laurel Avenue
Barangay San Antonio, Makati City, Metro Manila, Philippines

The fictional office is assumed to occupy a **single office floor** with approximately 20 employee work areas, a small server/network room, meeting space, reception, and common areas.

1.6 Organizational Structure

The organization uses a relatively flat startup structure.

```
Managing Director / CEO
 |
 ├──── Information Technology Manager
 |      ├──── Junior System Administrator
 |      ├──── Helpdesk / IT Support
 |      └──── Developers / Technical Staff
 |
 ├──── Human Resources Manager
```

│ └──── HR Staff

│

├──── Finance Manager

│ └──── Finance Staff

│

└──── Sales Manager

└──── Sales Staff

The **Junior System Administrator** works under the IT function and supports all departments. The IT team is responsible for endpoint deployment, network administration, server management, user support, backups, security configuration, and infrastructure documentation.

1.7 Employee Distribution

Department	Headcount	Primary IT Requirements
Information Technology	5	Developer workstations, laptops, virtualization, server/network administration
Human Resources	4	Office applications, HR records, secure document access
Finance	5	Office applications, spreadsheets, accounting/financial documents, secure storage
Sales	6	Office applications, CRM/web applications, presentations, mobile/laptop work
Total	**20**	**20 employees**

Part 2 — Enterprise Hardware Inventory

The design uses **14 desktops and 6 laptops**, exactly matching the 20 employees.

Asset ID	Hardware	Quantity	Department	Purpose
PC-IT-001–003	Business Desktop Computer	3	IT	Development, administration, testing, and general IT work
LAP-IT-001–002	Business Laptop	2	IT	Mobile administration, testing, meetings, and remote support

Asset ID	Hardware	Quantity	Department	Purpose
PC-HR-001–004	Business Desktop Computer	4	HR	HR records, Office applications, communications, and documentation
PC-FIN-001–005	Business Desktop Computer	5	Finance	Financial spreadsheets, accounting systems, reporting, and document processing
PC-SAL-001–002	Business Desktop Computer	2	Sales	Office work, CRM/web applications, presentations, and general sales administration
LAP-SAL-001–004	Business Laptop	4	Sales	Customer meetings, presentations, travel, remote work, and demonstrations
SRV-001	Rack/Tower Server	1	IT	File/application services, internal systems, monitoring, and infrastructure services
RTR-001	Business Router	1	IT	WAN routing and connection between ISP equipment and the security gateway
SW-001	48-Port Managed Gigabit Switch	1	IT	Central wired network connectivity and VLAN segmentation
PRN-001	Network Multifunction Printer	1	Shared	Printing/scanning for HR, Finance, Sales, and IT
UPS-001	Online UPS	1	IT	Protects server and core network equipment from power loss/surges
UPS-002	Line-Interactive UPS	1	Finance	Protects shared printer/critical Finance workstation equipment
UPS-003	Line-Interactive UPS	1	IT	Provides additional protection for network/communications equipment
AP-001–002	Enterprise Wireless Access Point	2	Shared/IT	Office Wi-Fi coverage and controlled wireless access
NAS-001	4-Bay NAS Storage	1	IT	Central backup repository and secondary file/storage location
BKP-001–002	External Backup Drive	2	IT	Offline/rotated backup copies
MON-IT-001–008	24-inch/27-inch Business Monitor	8	IT	Dual-monitor productivity for developers and administrators
MON-HR-001–004	Business Monitor	4	HR	Standard employee workstation displays

Asset ID	Hardware	Quantity	Department	Purpose
MON-FIN-001–006	Business Monitor	6	Finance	Dual-monitor capability for selected financial workloads
MON-SAL-001–006	Business Monitor	6	Sales	Standard workstation displays and presentation work

Hardware Quantity Summary

Hardware	Total
Desktop Computers	**14**
Laptops	**6**
Employee Computers	**20**
Server	**1**
Router	**1**
Managed Switch	**1**
Printer	**1**
UPS	**3**
Wireless Access Points	**2**
NAS	**1**
External Backup Drives	**2**
Monitors	**24**

Hardware Justification

Desktop computers: Desktops are appropriate for employees who primarily work at fixed office locations. HR and Finance have predictable desk-based workflows, while IT benefits from powerful development and administration workstations.

Laptops: Six laptops are assigned to IT and Sales. Sales has a higher mobility requirement because employees may meet customers, give presentations, and work outside the office. IT also requires mobile devices for infrastructure administration and troubleshooting.

Server: A dedicated server provides centralized infrastructure services without relying exclusively on employee computers or cloud services.

48-port managed switch: Although approximately 20 employee computers are initially deployed, a 48-port switch provides enough ports for the server, printer, APs, network devices, NAS, spare connections, and future employees.

NAS: The NAS provides centralized backup storage and can retain backup copies independently from the primary server.

UPS units: Power interruptions can corrupt data or abruptly shut down infrastructure. UPS protection is therefore important for the server and core network equipment.

Dual monitors: Developers and administrators benefit from multiple displays, while Finance employees frequently work with spreadsheets, reports, and source documents simultaneously.

Part 3 — Enterprise Software Inventory

The employee endpoint baseline is **Windows 11 Pro 25H2**. The server uses **Ubuntu Server 24.04.4 LTS**.

Software	Version	License	Purpose
Windows 11 Pro	25H2	Commercial/OEM or Volume License	Standard operating system for all 20 employee PCs
Ubuntu Server	24.04.4 LTS	Open Source / Ubuntu license	Server operating system for infrastructure and internal services
Microsoft 365 Apps / Office	Current Channel, Version 2606 baseline	Subscription	Word, Excel, PowerPoint, Outlook and productivity
Visual Studio Code	Current Stable	Free / Microsoft license	Source-code editing and development
Git	Current Stable	Open Source, GPLv2	Source-code version control
GitHub Desktop	Current Stable	Free	Simplifies Git/GitHub workflows for developers
VirtualBox	Current Stable	GPLv3 / Extension Pack subject to license	Local virtual machines for testing and system administration
Google Chrome	Current Stable	Free / Proprietary	Web applications, research, testing, and SaaS access

Software	Version	License	Purpose
Microsoft Defender	Windows-integrated / Defender for Business where licensed	Included / Subscription depending on deployment	Endpoint malware and threat protection
AnyDesk	Current Business-supported release	Commercial	Authorized remote support and troubleshooting
7-Zip	Current Stable	Open Source / LGPL	File compression and archive management

Microsoft 365 Apps is a subscription-based service that provides continuously updated versions of applications such as Word, Excel, and PowerPoint. Microsoft currently documents Version 2606 as a current supported build in the 2026 release cycle.

Software Justification

Windows 11 Pro: Provides the common endpoint platform. The Pro edition is appropriate for a business because it supports enterprise-oriented management and security capabilities such as BitLocker. Microsoft documents BitLocker support for Windows Pro editions.

Ubuntu Server: A cost-effective, stable Linux server platform suitable for web services, internal applications, development infrastructure, monitoring, and other server workloads. Ubuntu LTS releases receive long-term security maintenance.

Microsoft 365: HR, Finance, Sales, and IT all require professional document, spreadsheet, presentation, email, and collaboration capabilities.

VS Code: Appropriate for the development-oriented IT team because it supports many programming languages and development workflows.

Git: Essential for software development because it provides distributed version control.

GitHub Desktop: Makes Git operations easier for staff who prefer a graphical interface.

VirtualBox: Allows IT staff and developers to create isolated test environments without requiring additional physical computers.

Google Chrome: Supports web-based business systems, cloud applications, research, customer portals, and browser testing.

Microsoft Defender: Provides baseline endpoint security. For a 20-person organization, Defender for Business is also a reasonable upgrade because Microsoft positions it specifically for small and medium-sized organizations of up to 300 users.

AnyDesk: Allows IT personnel to provide remote support when properly authorized. Remote access should require strong authentication, logging, and restricted administrative permissions.

7-Zip: Provides efficient compression/decompression for development packages, backups, logs, and document archives.

Part 4 — Enterprise Network Inventory

Asset ID	Network Equipment	Quantity	Department/Area	Purpose
ISP-MDM-001	ISP Modem/ONT	1	Network Room	Terminates the ISP connection and provides the Ethernet WAN handoff
RTR-001	Business Router	1	Network Room	Routes the WAN connection from the ISP modem toward the security gateway
FW-001	Next-Generation Firewall	1	Network Room	Firewalling, NAT, VPN, traffic filtering, VLAN gateway/security policies
SW-001	48-Port Managed Gigabit Switch	1	Network Room	Central wired connectivity and VLAN segmentation
AP-001–002	Enterprise Wi-Fi 6 Access Point	2	Office Floor	Wireless coverage for staff and approved mobile devices
PP-001	48-Port CAT6 Patch Panel	1	Network Rack	Structured cable termination and cable management
CAB-CAT6-001–030	CAT6 Ethernet Cable Runs	30	Office	Permanent structured cabling from patch panel to outlets
RJ45-001–060	RJ45 Connectors	60	IT Stock	Termination/spare connectors for CAT6 patch cables and repairs

Recommended VLAN Design

VLAN	Name	Addressing Example	Purpose
10	IT	10.10.10.0/24	IT workstations and administration
20	HR	10.10.20.0/24	HR workstations
30	Finance	10.10.30.0/24	Finance workstations and financial systems
40	Sales	10.10.40.0/24	Sales workstations and laptops
50	Infrastructure	10.10.50.0/24	Server, NAS, printer, and network management

VLAN	Name	Addressing Example	Purpose
60	Corporate Wi-Fi	10.10.60.0/24	Employee wireless devices
70	Guest Wi-Fi	10.10.70.0/24	Internet-only guest access

The firewall should control traffic between these networks. For example:

- HR should not directly administer IT devices.
- Finance should have restricted access to sensitive systems.
- Guest Wi-Fi should have **Internet access only**.
- Infrastructure devices should be accessible only by authorized IT administrators.
- IT administrators may have controlled management access across the required VLANs.

Part 5 — Enterprise Network Diagram

5.1 Diagram Architecture

The recommended logical topology is:

Internet

|

▼

ISP Modem / ONT

|

▼

Router

|

▼

Next-Generation Firewall

```
 │
 ▼
48-Port Managed Switch
 ├──── VLAN 10 — IT
 ├──── VLAN 20 — HR
 ├──── VLAN 30 — Finance
 ├──── VLAN 40 — Sales
 ├──── VLAN 50 — Infrastructure
 │     ├──── Ubuntu Server
 │     ├──── NAS
 │     └──── Network Printer
 │
 ├──── AP-001
 │     ├──── Corporate Wi-Fi
 │     └──── Guest Wi-Fi
 │
 └──── AP-002
       ├──── Corporate Wi-Fi
       └──── Guest Wi-Fi
```

5.2 Nodes to Create in Draw.io

Node	Draw.io Symbol/Shape	Label
Internet	Cloud	**Internet — ISP WAN**
ISP Modem	Network/Modem device	**ISP Modem/ONT — ISP Handoff**
Router	Router/network device icon	**RTR-001 — WAN Router**
Firewall	Firewall/security appliance	**FW-001 — Next-Generation Firewall**
Switch	Network switch icon	**SW-001 — 48-Port Managed Switch**
AP-001	Wireless access point icon	**AP-001 — Wi-Fi 6 AP**
AP-002	Wireless access point icon	**AP-002 — Wi-Fi 6 AP**
Server	Server/rack server icon	**SRV-001 — Ubuntu Server 24.04.4 LTS**
NAS	Storage/NAS icon	**NAS-001 — Backup Storage**
Printer	Printer icon	**PRN-001 — Network MFP**
IT	Group/container or computer cluster	**VLAN 10 — IT — 10.10.10.0/24**
HR	Group/container or computer cluster	**VLAN 20 — HR — 10.10.20.0/24**
Finance	Group/container or computer cluster	**VLAN 30 — Finance — 10.10.30.0/24**
Sales	Group/container or computer cluster	**VLAN 40 — Sales — 10.10.40.0/24**
Infrastructure	Group/container	**VLAN 50 — Infrastructure — 10.10.50.0/24**
Corporate Wi-Fi	Wireless/group container	**VLAN 60 — Corporate Wi-Fi**
Guest Wi-Fi	Wireless/group container	**VLAN 70 — Guest Wi-Fi**

5.3 Exact Connections

Build the diagram in this order.

Connection 1 — Internet to ISP Modem

Internet → ISP Modem/ONT

Connection label: `ISP WAN / Fiber or Cable`

Connection 2 — ISP Modem to Router

ISP Modem/ONT → RTR-001**Connection label:** `WAN Ethernet`

Connection 3 — Router to Firewall

RTR-001 → FW-001

Connection label: `WAN / Transit Network`

The router provides the ISP-facing routing function, while the firewall becomes the primary security control for the internal network.

Connection 4 — Firewall to Switch

FW-001 → SW-001

Connection label: `802.1Q VLAN Trunk`

The firewall provides the default gateway/security policy for the internal VLANs.

Connection 5 — Switch to IT

SW-001 → IT

Port type: Access port
Label: `VLAN 10 – IT`

Connect the IT department's three desktops and two laptops through the appropriate wired/wireless infrastructure.

Connection 6 — Switch to HR

SW-001 → HR

Port type: Access port
Label: `VLAN 20 – HR`

Connection 7 — Switch to Finance

SW-001 → Finance

Port type: Access port
Label: `VLAN 30 — Finance`

Connection 8 — Switch to Sales

SW-001 → Sales

Port type: Access port
Label: `VLAN 40 — Sales`

Connection 9 — Switch to Infrastructure

SW-001 → Infrastructure

Port type: Access ports
Label: `VLAN 50 — Infrastructure`

Place the following devices inside the Infrastructure group:

SRV-001 — Ubuntu Server

NAS-001 — NAS Storage

PRN-001 — Network Printer

Connection 10 — Switch to AP-001

`SW-001 → AP-001`

Port type: VLAN trunk
Label: `802.1Q Trunk — VLAN 60 Corporate / VLAN 70 Guest`

Connection 11 — Switch to AP-002

`SW-001 → AP-002`

Port type: VLAN trunk
Label: `802.1Q Trunk — VLAN 60 Corporate / VLAN 70 Guest`

The APs broadcast two logical wireless networks:

- **ABC-Corporate** → VLAN 60
- **ABC-Guest** → VLAN 70

Guest traffic is isolated by the firewall and cannot access internal corporate VLANs.

5.4 Recommended Draw.io Layout

Use a **top-to-bottom network topology**:

```
         [ INTERNET ]
              |
        [ ISP MODEM/ONT ]
              |
         [ RTR-001 ]
              |
         [ FW-001 ]
              |
        [ SW-001 / 48P ]
   _________/ | | \_________
   /         | |         \
 [IT]      [HR] [Finance] [Sales]
   \
 [Infrastructure]
  /     |     \
[Server]  [NAS]  [Printer]


     / \
   [AP1][AP2]
```

The actual Draw.io diagram should expand the department groups to show their endpoint counts where space permits.

5.5 Diagram Title

Use:

ABC Startup Solutions — Enterprise Network Infrastructure Diagram

Under the title, add:

Student: [Your Name]

The final exported PNG/PDF from Draw.io should be referenced in the README as the official visual version of this specification.

Part 6 — System Administration Roles

6.1 Helpdesk Technician

Responsibilities

- Receive and resolve user support requests.
- Install and configure employee computers.
- Troubleshoot hardware and software problems.
- Create and maintain user accounts.
- Document incidents and resolutions.
- Escalate complex network/server problems.
- Assist with onboarding and offboarding.

Skills

- Windows troubleshooting
- Hardware diagnosis
- Basic networking
- Customer service
- Ticket management
- Basic cybersecurity awareness
- Documentation

Common Tools

- Windows administrative tools
- Microsoft 365 administration tools
- Remote support software such as AnyDesk
- Ticketing systems

- PowerShell
- Device Manager
- Event Viewer

Certifications

- CompTIA A+
- CompTIA Network+
- Microsoft fundamentals certifications
- ITIL Foundation

6.2 Network Administrator

Responsibilities

- Configure routers, switches, firewalls, and access points.
- Maintain VLANs and IP addressing.
- Monitor network availability and performance.
- Troubleshoot connectivity issues.
- Manage VPNs and firewall rules.
- Maintain network documentation.
- Implement network security controls.

Skills

- TCP/IP
- VLANs
- DHCP/DNS
- Routing and switching
- Wireless networking
- Firewall configuration
- Network monitoring
- Troubleshooting

Common Tools

- Cisco/other vendor CLI tools
- Firewall management consoles
- Wireshark
- Ping/traceroute
- SNMP monitoring platforms
- IP address management tools

Certifications

- CompTIA Network+

- Cisco CCNA
- Fortinet certification
- Juniper certifications

6.3 Linux System Administrator

Responsibilities

- Install and maintain Linux servers.
- Manage users, groups, permissions, and services.
- Apply security updates.
- Configure SSH and server networking.
- Manage storage and filesystems.
- Monitor system resources.
- Automate repetitive administrative tasks.
- Maintain server backups and recovery procedures.

Skills

- Linux command line
- Bash scripting
- Systemd
- SSH
- Networking
- Storage management
- Permissions
- Security hardening
- Automation

Common Tools

- Bash
- SSH
- `systemctl`
- `journalctl`
- `top`/`htop`
- `rsync`
- `cron`
- Git
- Ansible

Certifications

- Linux+
- LPIC-1/LPIC-2
- Red Hat Certified System Administrator

- Ubuntu-related professional training/certifications

6.4 Cloud Administrator

Responsibilities

- Manage cloud accounts and resources.
- Configure virtual machines and storage.
- Manage cloud identities and permissions.
- Monitor cloud resources.
- Configure backups and disaster recovery.
- Control cloud costs.
- Secure cloud infrastructure.
- Support hybrid/on-premises-to-cloud integration.

Skills

- Cloud networking
- Identity and access management
- Virtual machines
- Storage
- Monitoring
- Security
- Automation
- Infrastructure as Code

Common Tools

- Microsoft Azure portal/CLI
- AWS Management Console/CLI
- Google Cloud Console
- Terraform
- PowerShell
- Azure Monitor or equivalent cloud monitoring tools

Certifications

- Microsoft Azure Administrator Associate
- AWS Certified Cloud Practitioner
- AWS Certified Solutions Architect
- CompTIA Cloud+
- Google Associate Cloud Engineer

6.5 How the Four Roles Collaborate

At ABC Startup Solutions, the four roles form a support chain rather than operating independently. The **Helpdesk Technician** is normally the first point of contact when employees experience problems and resolves routine endpoint issues or escalates them. The **Network Administrator** handles connectivity, VLAN, firewall, Wi-Fi, and routing problems when a problem is network-related. The **Linux System Administrator** maintains the Ubuntu server, services, storage, logs, and server security. The **Cloud Administrator** manages cloud-hosted resources and assists when the company begins moving workloads or backups into the cloud. In a small startup, one person may perform several of these roles, so strong documentation, communication, change management, and escalation procedures are especially important.

Part 7 — Infrastructure Recommendations

7.1 Internet Provider Choice

ABC Startup Solutions should purchase a **business-grade fiber Internet connection**, rather than a residential plan.

Recommended baseline

- Minimum: **500 Mbps symmetrical**
- Preferred starting point: **1 Gbps symmetrical**, if affordable
- Business service-level agreement
- Static public IP option
- ISP technical support
- Installation of the ISP modem/ONT
- Option for a secondary ISP as the company grows

Reasoning

The company is a software development business. Developers may download large packages, synchronize repositories, use cloud services, participate in video meetings, and access remote infrastructure. Sales also needs reliable connectivity for customer presentations and cloud applications.

A symmetrical connection is particularly useful because the company may upload source code, backups, build artifacts, and other data.

7.2 Server Specifications

Recommended initial server:

Component	Recommendation
Form factor	Business tower or rack server
CPU	8–12 modern server-class cores
RAM	64 GB ECC RAM
OS storage	2 × 960 GB enterprise SSD in RAID 1
Data storage	4 × 2 TB enterprise SSD/HDD depending workload
RAID	Hardware RAID or software storage redundancy appropriate to server platform
Network	Dual 1 GbE or preferably 2.5/10 GbE capability
PSU	Redundant power supplies where budget allows
UPS	Connected to UPS-001
OS	Ubuntu Server 24.04.4 LTS
Management	Out-of-band management such as iDRAC/iLO/IPMI

Reasoning

A startup does not initially require an expensive enterprise cluster. However, the server should have enough CPU and memory for virtualization, internal services, monitoring, development services, and future applications.

ECC memory, redundant storage, and a UPS reduce the likelihood of hardware-related service interruption.

7.3 Backup Strategy

ABC Startup Solutions should implement a **3-2-1 backup strategy**:

3 copies of important data, on 2 different types of storage, with 1 copy kept offline/off-site.

Proposed implementation

Primary copy:

- Ubuntu Server/NAS production data

Secondary copy:

- NAS-001

Offline copy:

- BKP-001 and BKP-002 external drives, rotated regularly

Backup schedule

Data	Frequency
Critical company data	Daily
Server configuration	Daily
NAS snapshots	Daily
External offline backup	At least weekly
Full recovery test	Monthly
Disaster recovery exercise	Quarterly

The two external drives should be rotated so that one is stored offline while the other is being used for the latest backup.

CISA recommends automated/continuous backup of critical information and keeping an air-gapped/offline copy.

For additional resilience, the company should later add **encrypted cloud backup** as an off-site third copy.

7.4 Security Recommendations

Network security

- Deploy the dedicated next-generation firewall.
- Use VLAN segmentation.
- Block unnecessary inbound Internet traffic.
- Use outbound filtering where appropriate.
- Separate guest Wi-Fi from corporate networks.
- Disable unused switch ports.
- Use secure management protocols such as SSH/HTTPS.
- Change default device credentials.
- Keep firmware updated.

Endpoint security

- Enable Microsoft Defender.
- Enable BitLocker on Windows Pro endpoints.
- Enforce automatic security updates.
- Use standard user accounts for normal work.
- Restrict local administrator access.
- Enable screen locking.
- Maintain asset inventory.

BitLocker is specifically available on Windows Pro and provides protection against data exposure if a device is lost or stolen.

Identity security

- Require MFA.
- Use role-based access control.
- Apply least privilege.
- Disable accounts immediately during employee offboarding.
- Maintain separate administrator accounts.
- Audit privileged accounts.

CISA recommends MFA and least-privilege access as important security controls for small and medium-sized organizations.

7.5 Antivirus / Endpoint Protection

The baseline should use **Microsoft Defender** on all Windows 11 Pro systems.

For centralized business management, ABC Startup Solutions should consider **Microsoft Defender for Business**, especially if it adopts Microsoft 365 Business Premium. Microsoft describes Defender for Business as an endpoint security product designed for organizations with up to 300 users.

The policy should include:

- Real-time protection enabled
- Cloud-delivered protection enabled
- Automatic security intelligence updates
- Scheduled scanning
- Tamper protection where available
- Centralized alert monitoring
- Regular endpoint compliance reviews

7.6 Password Policy

The organization should use a modern password policy based on strong passphrases and MFA rather than forcing employees to change passwords on an arbitrary frequent schedule.

Recommended policy

- Minimum **14 characters** for normal employee passwords
- Prefer long passphrases
- Allow passwords/passphrases longer than 14 characters
- Block known compromised passwords
- Never reuse work passwords on personal websites
- Require MFA for cloud, VPN, administrator, and other high-risk accounts
- Use a company-approved password manager
- Lock or throttle repeated failed authentication attempts
- Never share administrator credentials
- Use separate privileged administrator accounts

NIST's current digital identity guidance is SP 800-63B-4, published in July 2025.

7.7 Expansion Plan

The infrastructure deliberately includes capacity beyond the initial 20 employees.

20 → 30 employees

- Add endpoint computers as required.
- Continue using the 48-port switch.
- Expand VLAN DHCP scopes.
- Add another AP if wireless capacity requires it.
- Increase NAS storage.
- Increase backup capacity.

30 → 50 employees

- Add a second managed switch.
- Upgrade uplinks if necessary.
- Consider 10 GbE between core infrastructure devices.
- Introduce centralized identity/device management.
- Add a secondary Internet connection.
- Consider a second server or virtualization host.
- Move selected services to cloud platforms.

50+ employees

- Introduce redundant switching/firewalls.
- Implement more formal network monitoring.
- Deploy centralized endpoint management.
- Consider a dedicated server/virtualization cluster.
- Implement a formal disaster recovery site/cloud environment.
- Expand the IT team into dedicated helpdesk, network, Linux, and cloud responsibilities.

This approach prevents ABC Startup Solutions from buying an oversized infrastructure on day one while still providing a practical path for growth.

Part 8 — Personal Reflection

As the newly hired Junior System Administrator for ABC Startup Solutions, this project taught me that system administration begins long before a computer is connected or a server is powered on. I learned that infrastructure planning requires me to understand the organization first, including its departments, employee numbers, business activities, and future goals. From there, I can determine what hardware, software, networking, security, and backup systems are actually necessary.

The most challenging task for me was making sure that all parts of the infrastructure plan remained consistent. For example, the number of computers had to match the 20 employees, while the network diagram needed to represent the same server, switch, access points, printer, and department segments listed in the inventories. I also had to think about how VLANs could separate IT, HR, Finance, and Sales while still allowing authorized communication with shared infrastructure. This showed me that an infrastructure plan is not simply a collection of equipment lists; every component has to fit into a larger technical design.

I also learned why planning matters before deployment. If an administrator purchases equipment without considering future growth, there may not be enough network ports, storage, wireless capacity, or IP addresses when the company expands. Security also needs to be considered from the beginning. Implementing firewall rules, VLANs, backups, MFA, endpoint protection, and appropriate password policies after a security incident would be much more difficult than building them into the infrastructure from the start.

This project helped me understand the relationship between different areas of system administration. Helpdesk support, networking, Linux administration, cloud administration, cybersecurity, and backup management are connected. A problem that appears to be a user's computer issue could actually be caused by DNS, a VLAN, a firewall rule, or a server service.

Most importantly, this project has helped me grow from simply thinking about individual computers to thinking about an entire IT environment. As a Junior System Administrator, I need to develop the habit of documenting decisions, considering risks, planning for growth, and testing systems before deployment. I now have a better understanding of how careful infrastructure planning can make an organization more secure, reliable, manageable, and prepared for future change.